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**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**NETWORK SECURITY (CYS360)**  
**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

|        |                            |       |                   |                         |                                         |
|--------|----------------------------|-------|-------------------|-------------------------|-----------------------------------------|
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| Score: | <u>          </u>          | Date: | <u>          </u> | Instructor's Signature: | <u>                                </u> |

## Lab # 5: To Configure Load Balancing Mode in Firewall Hot Standby

**Objective:** The objective of this lab is to configure the Load Balancing mode in Firewall Hot Standby.

**Duration:** 3 Hours

**Expected Outcomes:** By the end of this lab session, students will be able to:

1. Understand the basic principles of hot standby
2. Understand the load balancing mode
3. Master the configuration of Hot Standby using the CLI

### Rubrics

| LAB PERFORMANCE INDICATOR | SUBJECT KNOWLEDGE | DATA ANALYSIS AND INTERPRETATION | ABILITY TO CONDUCT EXPERIMENT |
|---------------------------|-------------------|----------------------------------|-------------------------------|
| SCORE                     |                   |                                  |                               |

*Note: Student performance will be assessed based on this rubric during lab sessions.*

### Scenario

As shown in [Figure 1](#), the service interfaces of the FWs work at Layer 3 and are directly connected to switches.

The FWs are expected to work in load balancing mode. Normally, both FW\_A and FW\_B forward traffic. If either FW fails, the other FW forwards all traffic to ensure service continuity.

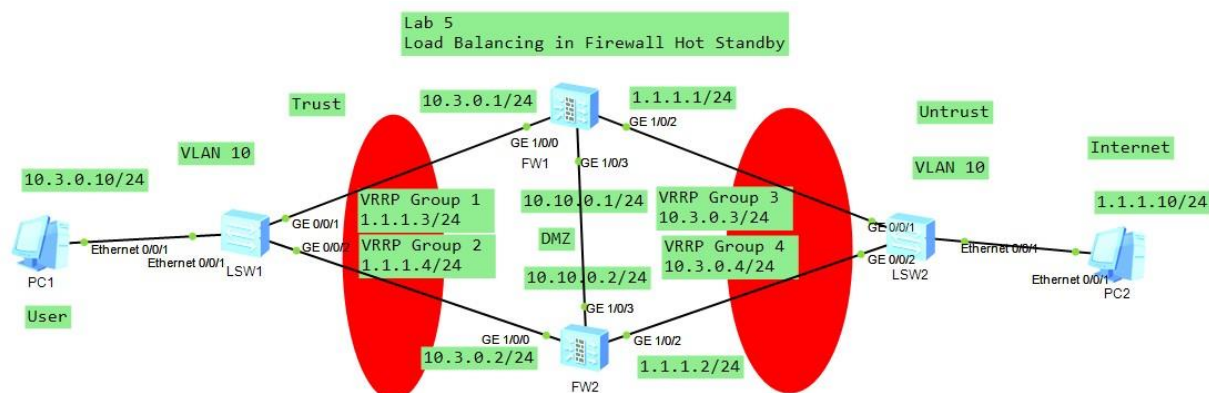


Figure 1 Firewall Hot Standby Load Balancing configuration

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## **CONFIGURATION**

**Complete basic network configurations.**

### **# Set IP addresses for the interfaces on FW\_A.**

```
<FW_A> system-view
[FW_A] interface GigabitEthernet 1/0/2
[FW_A-GigabitEthernet1/0/2] ip address 1.1.1.1 24
[FW_A-GigabitEthernet1/0/2] quit
[FW_A] interface GigabitEthernet 1/0/0
[FW_A-GigabitEthernet1/0/0] ip address 10.3.0.1 24
[FW_A-GigabitEthernet1/0/0] quit
[FW_A] interface GigabitEthernet 1/0/3
[FW_A-GigabitEthernet1/0/3] ip address 10.10.0.1 24
[FW_A-GigabitEthernet1/0/3] quit
```

### **# Set IP addresses for the interfaces on FW\_B.**

```
<FW_B> system-view
[FW_B] interface GigabitEthernet 1/0/2
[FW_B-GigabitEthernet1/0/2] ip address 1.1.1.2 24
[FW_B-GigabitEthernet1/0/2] quit
[FW_B] interface GigabitEthernet 1/0/0
[FW_B-GigabitEthernet1/0/0] ip address 10.3.0.2 24
[FW_B-GigabitEthernet1/0/0] quit
[FW_B] interface GigabitEthernet 1/0/3
[FW_B-GigabitEthernet1/0/3] ip address 10.10.0.2 24
[FW_B-GigabitEthernet1/0/3] quit
```

### **# Assign the interfaces to security zones on FW\_A.**

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface GigabitEthernet 1/0/0
[FW_A-zone-trust] quit
[FW_A] firewall zone dmz
[FW_A-zone-dmz] add interface GigabitEthernet 1/0/3
[FW_A-zone-dmz] quit
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface GigabitEthernet 1/0/2
[FW_A-zone-untrust] quit
```

### **# Assign the interfaces to security zones on FW\_B.**

```
FW_B] firewall zone trust
```

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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

```
[FW_B-zone-trust] add interface GigabitEthernet 1/0/0
[FW_B-zone-trust] quit
[FW_B] firewall zone dmz
[FW_B-zone-dmz] add interface GigabitEthernet 1/0/3
[FW_B-zone-dmz] quit
[FW_B] firewall zone untrust
[FW_B-zone-untrust] add interface GigabitEthernet 1/0/2
[FW_B-zone-untrust] quit
```

**# Create a default route with next hop 1.1.1.10 on FW\_A.**

```
[FW_A] ip route-static 0.0.0.0 0.0.0.0 1.1.1.10
```

**# Create a default route with next hop 1.1.1.10 on FW\_B.**

```
[FW_B] ip route-static 0.0.0.0 0.0.0.0 1.1.1.10
```

**Configure VRRP groups.**

**# To implement load balancing, configure two VRRP groups on each service interface and set the status of one VRRP group to Active and the other to Standby.**

**# Configure VRRP groups 1 and 2 on upstream service interface GE1/0/2 of FW\_A and set the status of VRRP group 1 to Active and status of VRRP group 2 to Standby.**

```
[FW_A] interface GigabitEthernet 1/0/2
[FW_A-GigabitEthernet1/0/2] vrrp vrid 1 virtual-ip 1.1.1.3 24 active
[FW_A-GigabitEthernet1/0/2] vrrp vrid 2 virtual-ip 1.1.1.4 24 standby
[FW_A-GigabitEthernet1/0/2] quit
```

**# Configure VRRP groups 3 and 4 on downstream service interface GE1/0/0 of FW\_A and set the status of VRRP group 3 to Active and status of VRRP group 4 to Standby.**

```
[FW_A] interface GigabitEthernet 1/0/0
[FW_A-GigabitEthernet1/0/0] vrrp vrid 3 virtual-ip 10.3.0.3 active
[FW_A-GigabitEthernet1/0/0] vrrp vrid 4 virtual-ip 10.3.0.4 standby
[FW_A-GigabitEthernet1/0/0] quit
```

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**NETWORK SECURITY (CYS360)**  
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**# Configure VRRP groups 1 and 2 on upstream service interface GE1/0/2 of FW\_B and set the status of VRRP group 1 to Standby and status of VRRP group 2 to Active.**

```
[FW_B] interface GigabitEthernet 1/0/2
[FW_B-GigabitEthernet1/0/2] vrrp vrid 1 virtual-ip 1.1.1.3 24 standby
[FW_B-GigabitEthernet1/0/2] vrrp vrid 2 virtual-ip 1.1.1.4 24 active
[FW_B-GigabitEthernet1/0/2] quit
```

**# Configure VRRP groups 3 and 4 on downstream service interface GE1/0/0 of FW\_B and set the status of VRRP group 3 to Standby and status of VRRP group 4 to Active.**

```
[FW_B] interface GigabitEthernet 1/0/0
[FW_B-GigabitEthernet1/0/0] vrrp vrid 3 virtual-ip 10.3.0.3 standby
[FW_B-GigabitEthernet1/0/0] vrrp vrid 4 virtual-ip 10.3.0.4 active
[FW_B-GigabitEthernet1/0/0] quit
```

**Configure quick session backup, specify the heartbeat interface, and enable hot standby.**

**# Configure quick session backup on both FWs in case of inconsistent forward and return packet paths.**

```
[FW_A] hrp mirror session enable
[FW_B] hrp mirror session enable
```

**# Specify the heartbeat interface and enable hot standby on FW\_A.**

```
[FW_A] hrp interface GigabitEthernet 1/0/3 remote 10.10.0.2
[FW_A] hrp enable
```

**# Specify the heartbeat interface and enable hot standby on FW\_B.**

```
[FW_B] hrp interface GigabitEthernet 1/0/3 remote 10.10.0.1
[FW_B] hrp enable
```

**Create a security policy on FW\_A. After hot standby relationship is established, the security policy on FW\_A will be automatically backed up to FW\_B.**

**# Configure a security policy to allow intranet users to access the Internet.**

```
HRP_M[FW_A] security-policy
HRP_M[FW_A-policy-security] rule name trust_to_untrust
HRP_M[FW_A-policy-security-rule-trust_to_untrust] source-zone trust
HRP_M[FW_A-policy-security-rule-trust_to_untrust] destination-zone untrust
```

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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

```
HRP_M[FW_A-policy-security-rule-trust_to_untrust] action permit
HRP_M[FW_A-policy-security-rule-trust_to_untrust] source-address 10.3.0.0 24
HRP_M[FW_A-policy-security-rule-trust_to_untrust] quit
HRP_M[FW_A-policy-security] quit
```

**Configure a NAT policy on FW\_A. After hot standby relationship is established, the NAT policy on FW\_A will be automatically backed up to FW\_B.**

**# Configure a NAT policy to translate source addresses on subnet 10.3.0.0/16 to an IP address in the NAT address pool (1.1.1.5 to 1.1.1.8) when intranet users access the Internet.**

```
HRP_M[FW_A] nat address-group group1
HRP_M[FW_A-address-group-group1] section 0 1.1.1.5 1.1.1.8
HRP_M[FW_A-address-group-group1] quit
HRP_M[FW_A] nat-policy
HRP_M[FW_A-policy-nat] rule name policy_nat1
HRP_M[FW_A-policy-nat-rule-policy_nat1] source-zone trust
HRP_M[FW_A-policy-nat-rule-policy_nat1] destination-zone untrust
HRP_M[FW_A-policy-nat-rule-policy_nat1] source-address 10.3.0.0 16
HRP_M[FW_A-policy-nat-rule-policy_nat1] action source-nat address-group group1
HRP_M[FW_A-policy-nat-rule-policy_nat1] quit
HRP_M[FW_A-policy-nat] quit
```

**# To prevent port conflicts in address translation on the FWs in load balancing mode, configure available port ranges respectively on FW\_A and FW\_B. The configuration on FW\_A is as follows:**

```
HRP_M[FW_A] hrp nat resource primary-group
```

**After the command is executed on FW\_A, FW\_B will automatically back it up and convert it to the hrp nat resource secondary-group command.**

**Configure the switches and PCs.**

**# Add the three interfaces of the switches to the same VLANs accordingly. For configuration commands, refer to related documents of the switches.**

**# On some intranet PCs, specify the IP address (10.3.0.3) of VRRP group 3 as the default gateway address and on some other intranet PCs, specify the IP address (10.3.0.4) of VRRP group 4 as the default gateway address to implement load balancing of intranet traffic.**

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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

## Verification

Run the **display vrrp** command on FW\_A to check the status information about the interfaces in the VRRP group. If the following information is displayed, the VRRP group is successfully created.

```
HRP_M<FW_A> display vrrp
GigabitEthernet1/0/1 | Virtual Router 1
  State : Master
  Virtual IP : 1.1.1.3
  Master IP : 1.1.1.1
  PriorityRun : 120
  PriorityConfig : 100
  MasterPriority : 120
  Preempt : YES   Delay Time : 0 s
  TimerRun : 60 s
  TimerConfig : 60 s
  Auth type : NONE
  Virtual MAC : 0000-5e00-0101
  Check TTL : YES
  Config type : vgmpp-vrrp
  Backup-forward : disabled
  Create time : 2015-03-17 17:35:54 UTC+08:00
  Last change time : 2015-03-22 16:01:56 UTC+08:00
GigabitEthernet1/0/1 | Virtual Router 2
  State : Backup
  Virtual IP : 1.1.1.4
  Master IP : 1.1.1.1
  PriorityRun : 120
  PriorityConfig : 100
  MasterPriority : 120
  Preempt : YES   Delay Time : 0 s
  TimerRun : 60 s
  TimerConfig : 60 s
  Auth type : NONE
  Virtual MAC : 0000-5e00-0102
  Check TTL : YES
  Config type : vgmpp-vrrp
  Backup-forward : disabled
```

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**NETWORK SECURITY (CYS360)**  
**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

```
Create time : 2015-03-17 17:35:54 UTC+08:01
Last change time : 2015-03-22 16:01:56 UTC+08:01
GigabitEthernet1/0/3 | Virtual Router 3
State : Master
Virtual IP : 10.3.0.3
Master IP : 10.3.0.1
PriorityRun : 120
PriorityConfig : 100
MasterPriority : 120
Preempt : YES   Delay Time : 0 s
TimerRun : 60 s
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0103
Check TTL : YES
Config type : vgmpp-vrrp
Backup-forward : disabled
Create time : 2015-03-17 17:35:54 UTC+08:02
Last change time : 2015-03-22 16:01:56 UTC+08:02
GigabitEthernet1/0/3 | Virtual Router 4
State : Backup
Virtual IP : 10.3.0.4
Master IP : 10.3.0.1
PriorityRun : 120
PriorityConfig : 100
MasterPriority : 120
Preempt : YES   Delay Time : 0 s
TimerRun : 60 s
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0104
Check TTL : YES
Config type : vgmpp-vrrp
Backup-forward : disabled
Create time : 2015-03-17 17:35:54 UTC+08:03
Last change time : 2015-03-22 16:01:56 UTC+08:03
```

Run the **display hrp state verbose** command on FW\_A to check the VGMP group status. If the following information is displayed, hot standby relationship is successfully established.

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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

```
HRP_M<FW_A> display hrp state verbose
Role: active, peer: active
Running priority: 46004, peer: 46004
Backup channel usage: 30%
Stable time: 1 days, 13 hours, 35 minutes
Last state change information: 2015-03-22 16:01:56 HRP core state changed, old_
state = normal(active), new_state = normal(active), local_priority = 46004,
peer_priority = 4604.

Configuration:
hello interval:          1000ms
preempt:                60s
mirror configuration:    off
mirror session:          off
track trunk member:      on
auto-sync configuration: on
auto-sync connection-status: on
adjust ospf-cost:        on
adjust ospfv3-cost:      on
adjust bgp-cost:         on
nat resource:            off
Detail information:
    GigabitEthernet1/0/1 vrrp vrid 1: active
    GigabitEthernet1/0/1 vrrp vrid 2: standby
GigabitEthernet1/0/3 vrrp vrid 3: active
    GigabitEthernet1/0/3 vrrp vrid 4: standby
```

### Questions:

1. Describe the Load Balancing Mode in Firewall Hot Standby. How it works and how it does load balancing?

Load Balancing Mode in Firewall Hot Standby means using more than one firewall at the same time to share network traffic.

How it works:

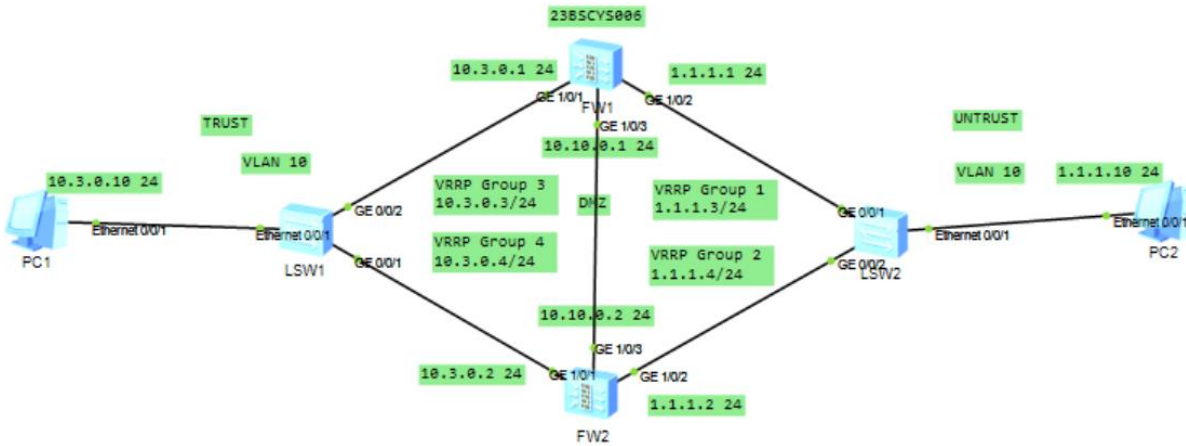
- Two or more firewalls are active.
- Traffic is split between them instead of using only one firewall.
- If one firewall fails, the other keeps working.

How it does load balancing:

- Network traffic is divided evenly between firewalls.
- Each firewall handles part of the traffic.
- This improves speed, performance, and reliability.



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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**



```

2026-01-19 19:09:10.300
Total:4 Master:2 Backup:2 Non-active:0
VRID State Interface Type Virtual IP
-----
1 Backup GE1/0/2 Vgmp 1.1.1.3
2 Master GE1/0/2 Vgmp 1.1.1.4
3 Backup GE1/0/1 Vgmp 10.3.0.3
4 Master GE1/0/1 Vgmp 10.3.0.4
RP_S[FW-2-23BSCYS006]
2026-01-19 19:09:40.730
Total:4 Master:2 Backup:2 Non-active:0
VRID State Interface Type Virtual IP
-----
1 Master GE1/0/2 Vgmp 1.1.1.3
2 Backup GE1/0/2 Vgmp 1.1.1.4
3 Master GE1/0/1 Vgmp 10.3.0.3
4 Backup GE1/0/1 Vgmp 10.3.0.4
HRP_M[FW-2-23BSCYS006]

```

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**NETWORK SECURITY (CYS360)**  
**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**

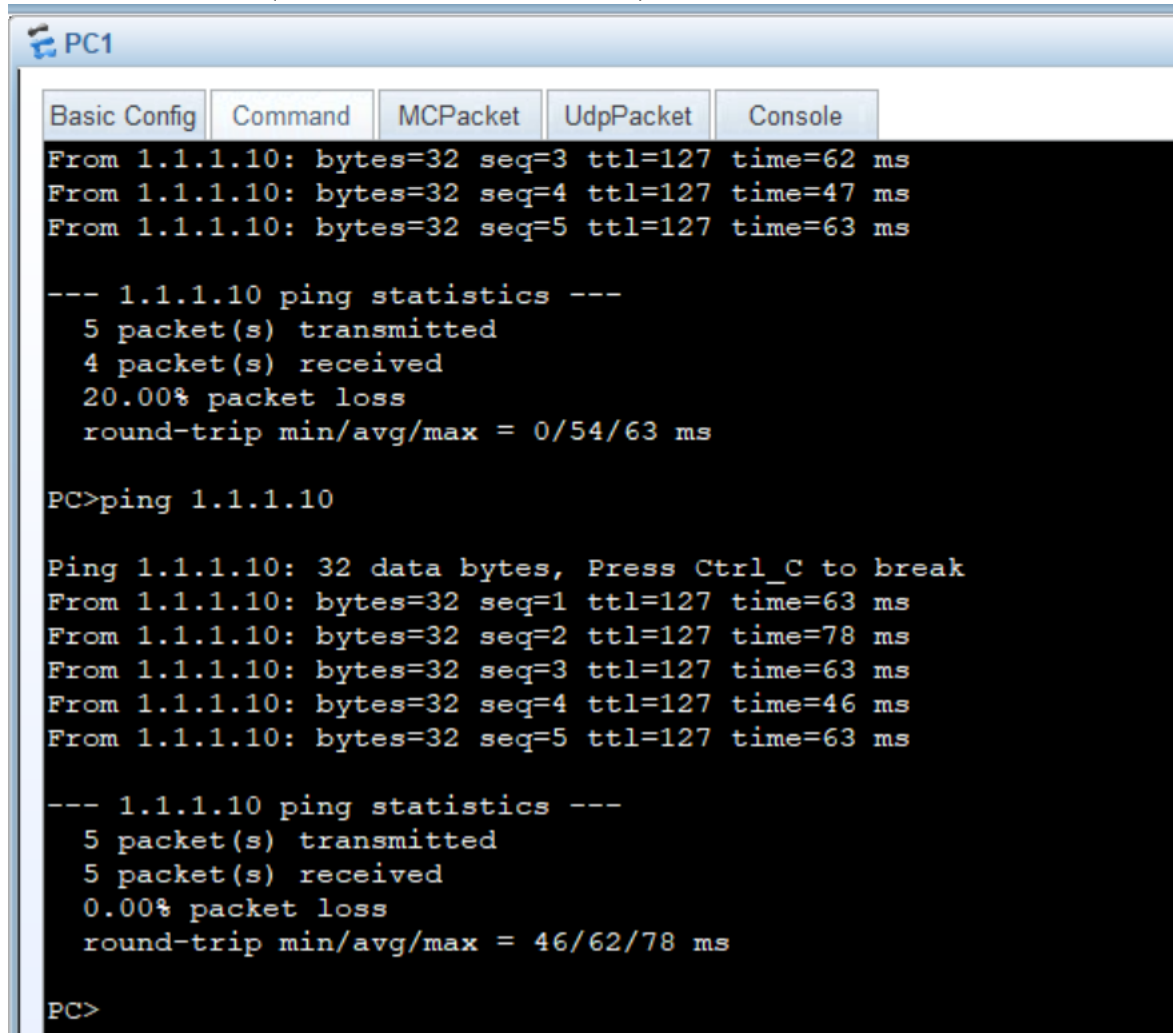
```
HRP_M[FW-2-23BSCYS006]dis hrp st
HRP_M[FW-2-23BSCYS006]dis hrp stat
HRP_M[FW-2-23BSCYS006]dis hrp state ve
HRP_M[FW-2-23BSCYS006]dis hrp state verbose
2026-01-19 19:13:16.060
Role: active, peer: active
Running priority: 45000, peer: 45000
Backup channel usage: 0.00%
Stable time: 0 days, 0 hours, 19 minutes
Last state change information: 2026-01-19 18:53:29 HRP core state changed, old_
state = abnormal(standby), new_state = normal, local_priority = 45000, peer_prio
rity = 45000.

Configuration:
hello interval:          1000ms
preempt:                 60s
mirror configuration:    off
mirror session:          on
track trunk member:      on
auto-sync configuration: on
auto-sync connection-status: on
adjust ospf-cost:        on
adjust ospfv3-cost:      on
adjust bgp-cost:         on
nat resource:            primary

Detail information:
    GigabitEthernet1/0/2 vrrp vrid 1: active
    GigabitEthernet1/0/2 vrrp vrid 2: standby
    GigabitEthernet1/0/1 vrrp vrid 3: active
    GigabitEthernet1/0/1 vrrp vrid 4: standby
                                ospf-cost: +0
                                ospfv3-cost: +0
                                bgp-cost: +0

HRP_M[FW-2-23BSCYS006]
HRP_M[FW-2-23BSCYS006]
HRP_M[FW-2-23BSCYS006]
```

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**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 5**



The screenshot shows a PC1 console window with a blue title bar and a tabbed interface. The 'Console' tab is selected, displaying a series of ping commands and their results. The first three lines show individual ping results from 1.1.1.10 to 1.1.1.10. The fourth line shows the ping statistics for 1.1.1.10, indicating a 20.00% packet loss. The fifth line shows the command 'PC>ping 1.1.1.10'. The sixth line shows the ping command being executed. The seventh line shows the ping results for 1.1.1.10, indicating a 0.00% packet loss. The eighth line shows the ping statistics for 1.1.1.10, indicating a 0.00% packet loss. The ninth line shows the command 'PC>'.

```
PC1
Basic Config Command MCPacket UdpPacket Console
From 1.1.1.10: bytes=32 seq=3 ttl=127 time=62 ms
From 1.1.1.10: bytes=32 seq=4 ttl=127 time=47 ms
From 1.1.1.10: bytes=32 seq=5 ttl=127 time=63 ms

--- 1.1.1.10 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/54/63 ms

PC>ping 1.1.1.10

Ping 1.1.1.10: 32 data bytes, Press Ctrl_C to break
From 1.1.1.10: bytes=32 seq=1 ttl=127 time=63 ms
From 1.1.1.10: bytes=32 seq=2 ttl=127 time=78 ms
From 1.1.1.10: bytes=32 seq=3 ttl=127 time=63 ms
From 1.1.1.10: bytes=32 seq=4 ttl=127 time=46 ms
From 1.1.1.10: bytes=32 seq=5 ttl=127 time=63 ms

--- 1.1.1.10 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 46/62/78 ms

PC>
```